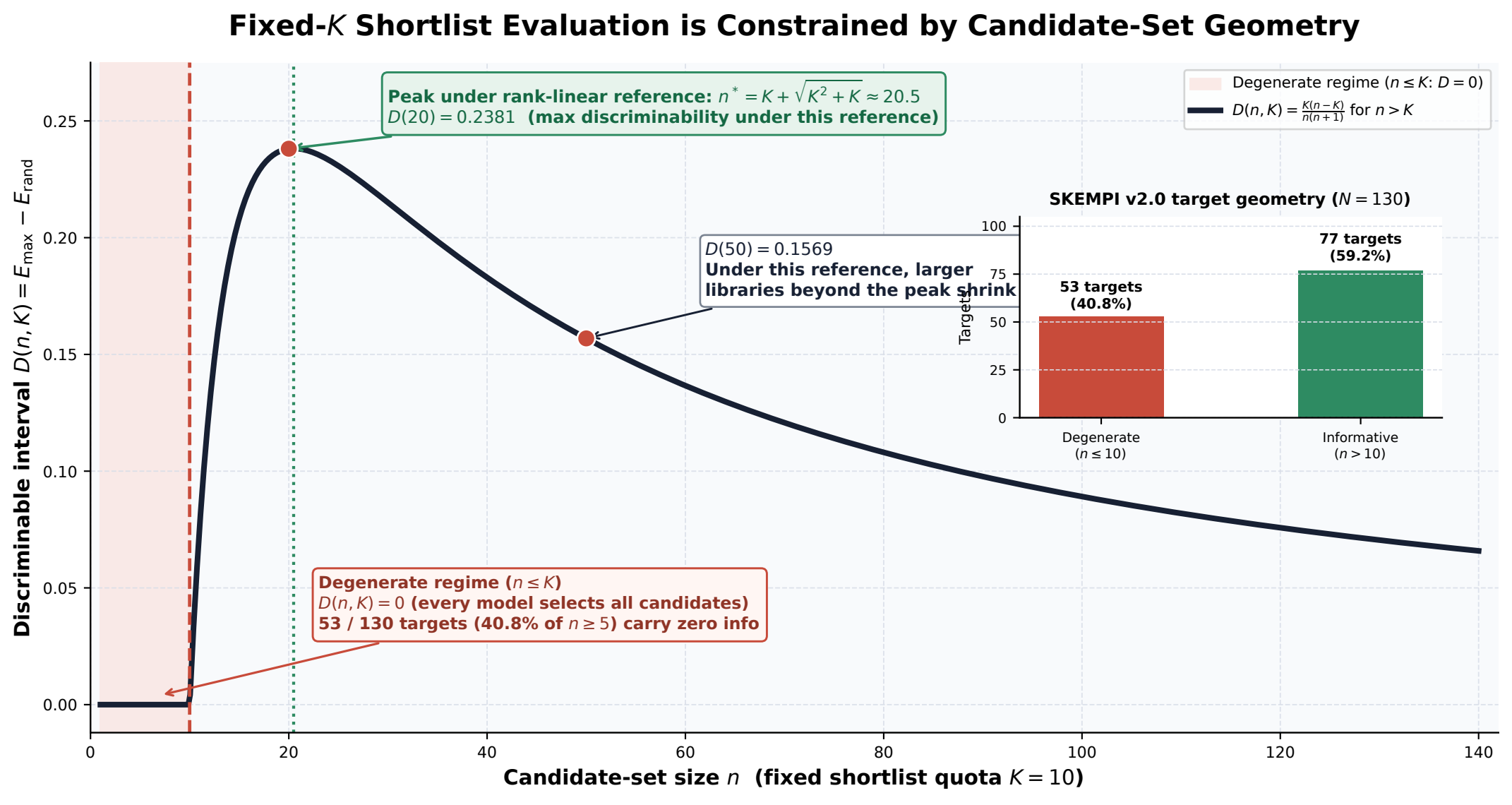
Fixed-K Shortlist Evaluation is Constrained by Candidate-Set Geometry
Degenerate regime ($n \le K$: $D = 0$)
$D(n,K) = \frac{K(n-K)}{n(n+1)}$ for $n > K$
Peak under rank-linear reference: $n^* = K + \sqrt{K^2 + K} \approx 20.5$
$D(20) = 0.2381$ (max discriminability under this reference)
$D(50) = 0.1569$
Under this reference, larger libraries beyond the peak shrink
Degenerate regime ($n \le K$)
$D(n,K) = 0$ (every model selects all candidates)
53 / 130 targets (40.8% of $n \ge 5$) carry zero info
SKEMPI v2.0 target geometry ($N = 130$)
53 targets (40.8%)
77 targets (59.2%)
Targets
Degenerate ($n \le 10$)
Informative ($n > 10$)
Discriminable interval $D(n,K) = E_{max} - E_{rand}$
Candidate-set size n (fixed shortlist quota $K = 10$)